An Internship Report
Of
CodeAlpha Virtual Internship Program in Full Stack Development

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B. Tech (Computer Science and Engineering)

By

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Section: 2CSE36
Batch: 2025-2029

CANDIDATE'S DECLARATION

I, Prabhat Kumar Jha, do hereby declare that the internship report titled "CodeAlpha Virtual Internship Program in Full Stack Development" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the CodeAlpha team, specifically Swati Srivastava, Founder & CEO, for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Full Stack Development. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Student Name: Prabhat Kumar Jha
Roll No.: 25SCS1003004567

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3. PROJECT DESCRIPTION

3.1 Introduction

This report provides a comprehensive account of the 4-week Virtual Internship undertaken in the domain of "Full Stack Development" with CodeAlpha. The internship program was completed from 10th July 2026 to 10th August 2026 as partial fulfillment of the academic requirements for the degree of Bachelor of Technology in Computer Science and Engineering (Section: 2CSE36) at the School of Computer Science and Engineering, IILM University, Greater Noida.

The primary objective of this internship was to translate theoretical software engineering and computer science concepts into practical, industry-standard web development practices. During the internship, the curriculum focused on mastering the complete software development lifecycle (SDLC), responsive frontend design, server-side routing, and relational database management.

As part of the practical deliverables, two major end-to-end full-stack applications were developed from scratch:

1. E-Commerce Store: A responsive digital storefront featuring dynamic product catalogs, asynchronous client-side API communication (api.js), user session authentication, real-time shopping cart management, and a persistent SQLite database (store.db).

2. CampusLoop Social Platform: A modern community platform tailored for student interaction, featuring a centralized feed interface, real-time search capabilities across posts and circles, dynamic user profile avatars, and backend database integration (campusloop.db).

The development workflow was executed in Visual Studio Code, utilizing Git for version control. The tech stack encompassed HTML5, CSS3, modern JavaScript (ES6+), Node.js, Express.js, and SQLite. Through these structured tasks, extensive hands-on expertise was gained in writing asynchronous functions (async/await), designing RESTful API endpoints, managing multi-tier directory architectures, and implementing secure middleware workflows.

3.2 Organization Profile

CodeAlpha is a software development organization recognized by the Ministry of Micro, Small & Medium Enterprises (MSME), Government of India. Operating under the motto "Your Dream Our Passion," CodeAlpha offers structured learning combined with practical application. The organization is led by Founder & CEO Swati Srivastava and focuses on equipping students with industry-relevant technical skills and hands-on project experience.

3.3 Problem Statement

While academic coursework provides the theoretical foundations of software engineering, students require structured, hands-on exposure to build complete, functional web applications from scratch. The core problem addressed during this internship was to bridge the gap between frontend design and backend API integration. This required the practical implementation of full-stack workflows—from designing responsive user interfaces and managing user authentication to handling database architectures for complex domains like e-commerce transactions and social media feeds.

3.4 Project Objectives

- To understand the first principles of full-stack development and RESTful API integration.
- To design and deploy a complete E-commerce Store featuring product listings, secure user authentication, and dynamic cart management.
- To architect a functional Social Media Platform named "CampusLoop" with interactive feeds, search capabilities, and user profile management.
- To gain hands-on experience using modern development environments, specifically Visual Studio Code, for managing project directories, routing, and middleware.

3.5 Scope of the Project

The scope of this internship spanned two distinct full-stack development tasks:

- **Task 1 (E-commerce Store):** The scope involved developing the client-side logic (`api.js`, `cart.js`, `product.js`) to handle asynchronous fetch requests to a backend API (`/api/auth/me`, `/api/cart`). It included building the user interface (`index.html`, `cart.html`, `orders.html`) and configuring the backend data store using SQLite (`store.db`).
- **Task 2 (Social Media Platform):** The scope centered on building "CampusLoop," a platform designed for students. It involved creating a premium navigation interface with a real-time search bar for posts and circles, integrating dynamic user avatars, and establishing the database architecture (`campusloop.db`).

3.6 Technologies and Tools Used

- **Frontend Technologies:** HTML5, CSS3, JavaScript (ES6+), DOM Manipulation.
- **Backend & API:** Node.js, Express.js (middleware and routing configuration).
- **Database:** SQLite (`store.db` and `campusloop.db`).
- **Development Tools:** Visual Studio Code (IDE), Git for version control (`.git` configurations).

3.7 System Architecture

The architecture implemented during both tasks followed a standard Client-Server model:

- **Client Layer:** Built with HTML/CSS and vanilla JavaScript. For the e-commerce store, `api.js` serves as the core utility for making authenticated asynchronous requests to the backend using the Fetch API.
- **Middleware Layer:** Express routing (`authRoutes.js`, `cartRoutes.js`, `productRoutes.js`) processes incoming HTTP requests and handles session credentials.
- **Database Layer:** Local SQLite databases (`store.db`, `campusloop.db`) maintain persistent records for users, products, cart items, and social media posts.

3.8 Methodology
The internship followed a project-based methodology emphasizing hands-on application:

Task	Project	Description
Task 1	E-commerce Store	Developed a full-stack digital storefront. Implemented dynamic cart updates reducing array quantities, secure user credential inclusion in headers, and API error handling.
Task 2	CampusLoop Social Platform	Built a community feed interface. Structured the HTML layout for search inputs, user greetings, and session logout functionality alongside backend server configurations.

3.9 Expected Outcomes

- Practical capability to manage multi-tiered directory structures (public assets, server routes, database files) in a professional IDE.
- Demonstrated ability to write asynchronous JavaScript functions (`async/await`) for seamless backend data retrieval.
- Successful implementation of responsive UI components, such as search bars and dynamic navigation links.
- Completion of all internship deliverables resulting in the award of the official CodeAlpha Virtual Internship Certificate.

4. PROJECT PROOFS

Figure 1: VS Code interface displaying the backend API configuration (api.js) and directory structure for the E-commerce Store project.

Figure 2: VS Code interface displaying the frontend structure (feed.html) and database setup for the CampusLoop Social Media Platform.

5. Internship Certificate

CODE ALPHA

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सत्यमेव जयते

MINISTRY OF CORPORATE AFFAIRS

GOVERNMENT OF INDIA

CERTIFICATE

OF COMPLETION

This Certificate Is Proudly Presented To

Prabhat Kumar Jha

This Certificate recognizes the successful completion of the CodeAlpha Virtual Internship Program in Full Stack Development, demonstrating dedication and sincerity throughout the program.

We appreciate the valuable efforts and wish continued success in all future endeavors.

10th July 2026 to 10th August 2026

CEO & Co-Founder

25th August 2026

Date of Issue

SCAN TO VERIFY THIS CERTIFICATE

Student ID: CA/DF1/194729

6. Offer Letter

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INTERNSHIP OFFER LETTER

7th July 2026 **STUDENT ID: CA/DF1/194729**

Name: Prabhat Kumar Jha

Welcome to CodeAlpha

Congratulations! We are delighted to make you an offer as **Full Stack Development internship** offer letter is to confirm your selection **Full Stack Development internship** at **CodeAlpha** and welcome you for the same effective from 10th July 2026 to 10th August 2026. Hope you will be doing well.

This Internship is observed by CodeAlpha as being a learning opportunity for you.

Your internship will embrace orientation and give emphasis on learning new skills with a deeper understanding of concepts through hands-on application of the knowledge you gained as an intern. You will acknowledge your obligation to perform all work allocated to you to the best of your ability within lawful and reasonable direction given to you.

We look forward to a worthwhile and fruitful association which will make you equipped for future projects.

Wishing you the most enjoyable and truly meaningful internship program experience.

Sincerely,

Swati Srivastava
Founder & CEO

CODEALPHA SOFTWARE DEVELOPMENT

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7. BIBLIOGRAPHY/REFERENCES

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2. CodeAlpha, "Certificate of Completion - Full Stack Development," Student ID: CA/DF1/194729, August 2026.

3. E-commerce Store Source Code & Project Files, August 2026.

4. CampusLoop Social Platform Source Code & Project Files, August 2026.